Assignment 4

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Course : MCA
Subject : Cyber Security
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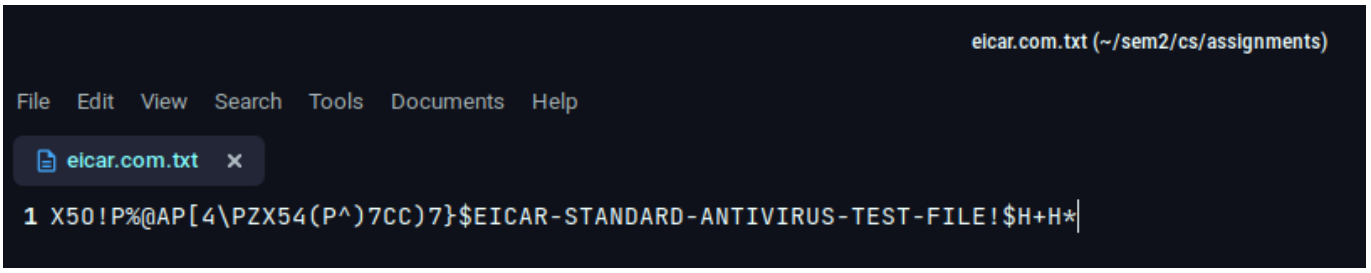
Task: Static Malware Analysis using Free Online Tools

1. Tool Setup & File Acquisition: Download the EICAR test file from Download Anti Malware Testfile - EICAR. Save it to your Downloads folder as eicar.com.txt. Confirm your operating system and antivirus settings allow temporary access to the file.




```
text got from eicar website, saved in a text file
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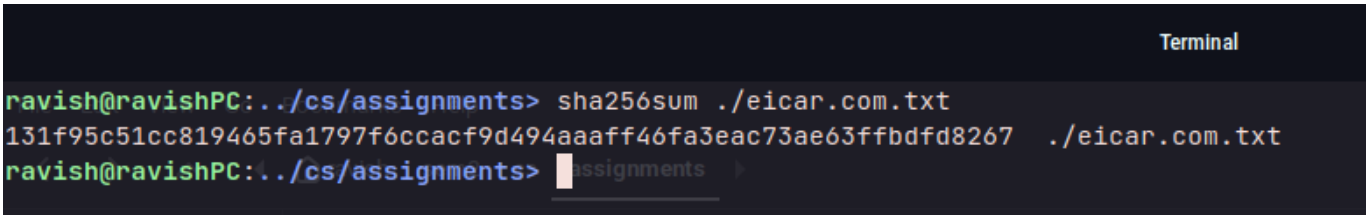
2. Hash Computation: Compute the SHA256 hash of the file using PowerShell on Windows or sha256sum on Linux/macOS. Record the output exactly as shown in your terminal.




```
sha256sum hased text :
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267
```

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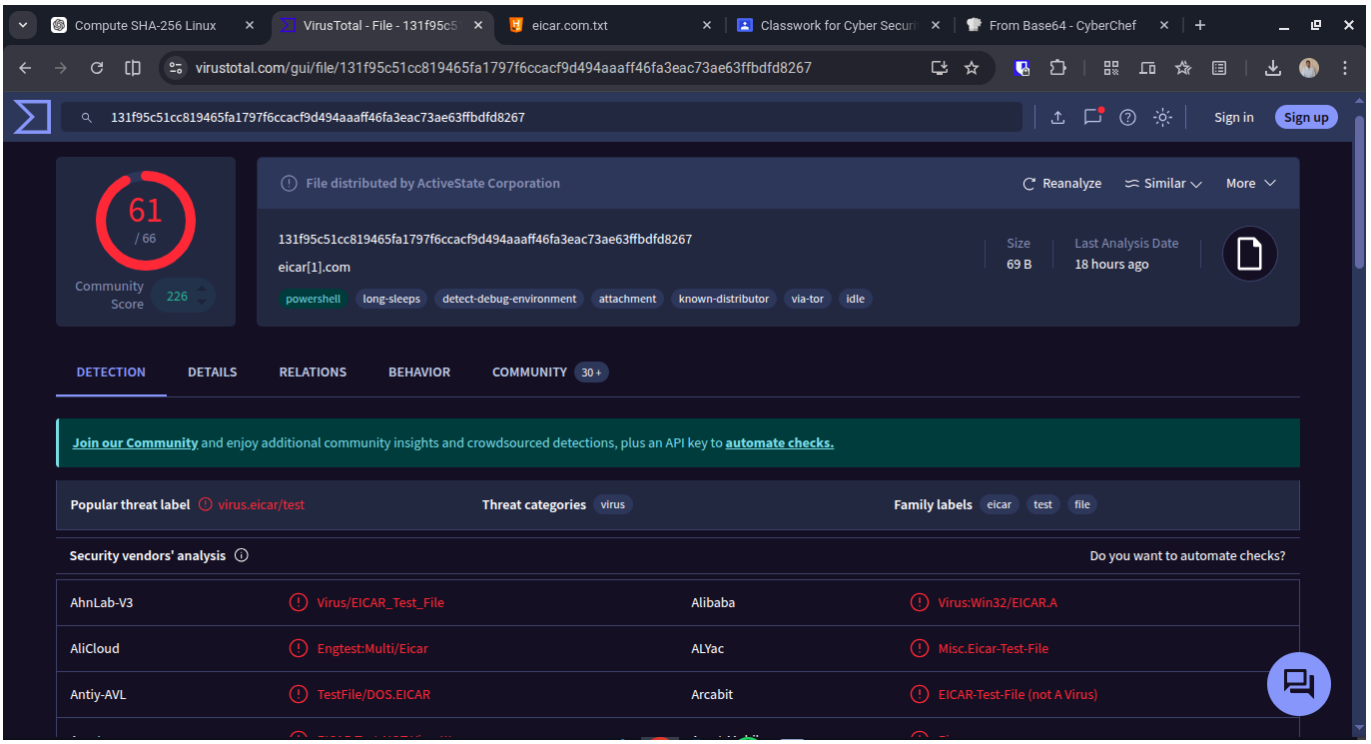
3. Multi-Engine Scan: Submit the SHA256 hash (not the file) to VirusTotal at VirusTotal. Examine the Detection tab and document the number of engines that flag the file as malicious, along with three named detections.



analysis site

```
No. of engines flagging malicious : 61
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Detector	Label
Alibaba	Virus:Win32/EICAR.A
Avast	EICAR Test-NOT Virus!!!
BitDefender	EICAR-Test-File (not A Virus)

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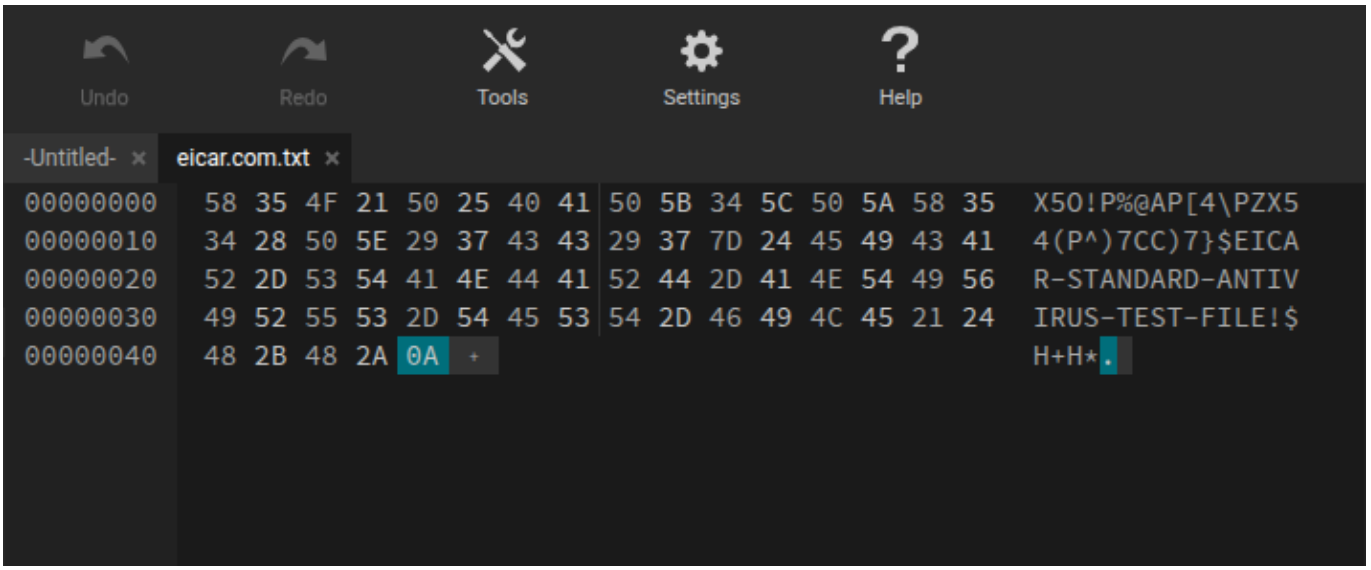
4. String & Hex Inspection: Open the file in Hexed.it (HexEd.it) and observe both the raw hexadecimal bytes and the ASCII representation. Identify and record all readable strings.



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Readable strings
    1. EICAR-STANDARD-ANTIVIRUS-TEST-FILE
```

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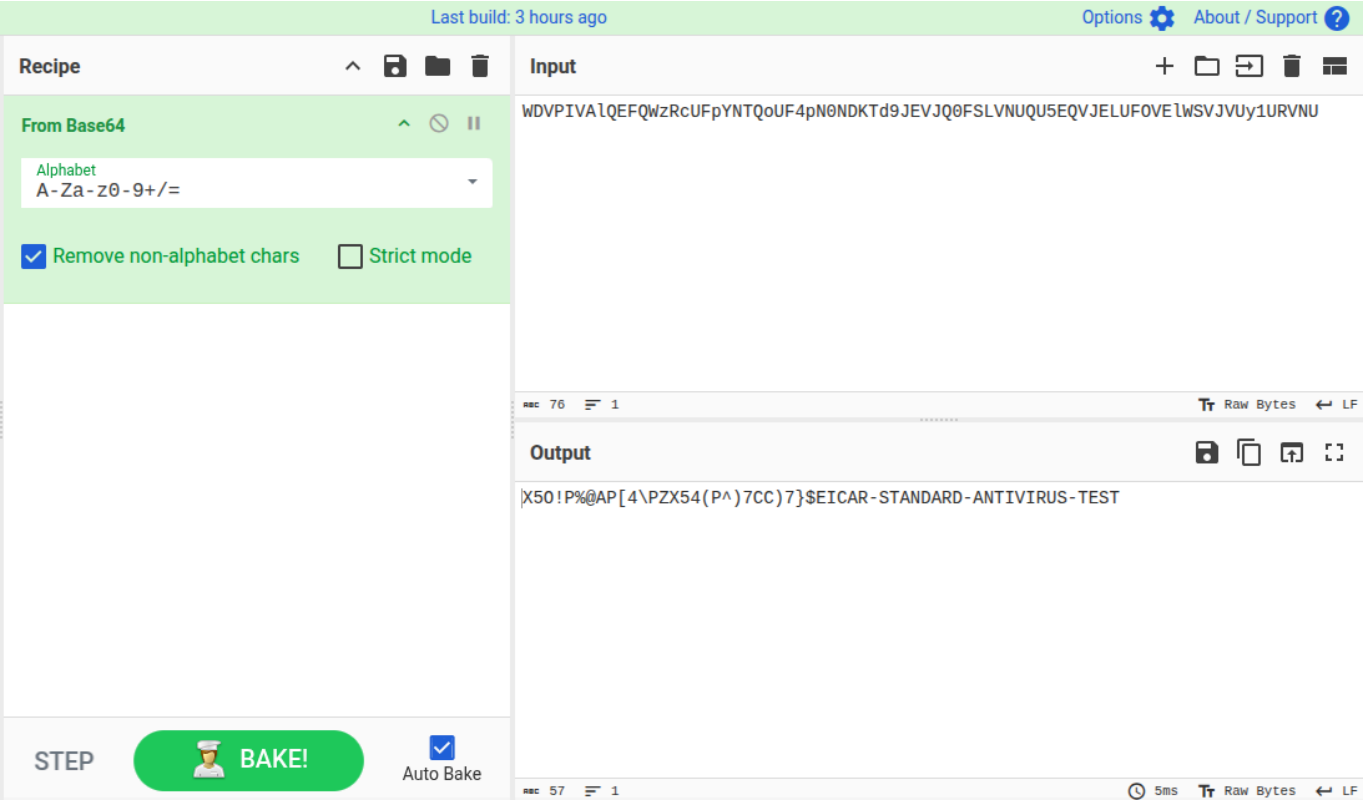
5. Decoding Practice: Use CyberChef (CyberChef) with the From Base64 operation to decode the following sample string and observe the output:
   `WDVPIVAlQEFQWzRcUFpYNTQoUF4pN0NDKTd9JEVJQ0FSLVNUQU5EQVJELUFOVElWSVJVUy1URVNU`




```
Input :
WDVPIVAlQEFQWzRcUFpYNTQoUF4pN0NDKTd9JEVJQ0FSLVNUQU5EQVJELUFOVElWSVJVUy1URVN
U

Output : X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST
```

Activity	what have i done	tools used	input	output
Checking file hash	used sha256sum on linux to get hash	sha256sum	eicar malware text	eicar text sha256sum hash
Check Antivirus Detection	scanned hash on virustotal website	virustotal.com	eicar text sha256sum hash	61 engines flagged Answer 3
Extract Strings	Viewed ASCII strings	Hexed.it	given encodede text	EICAR string found Answer 4
Hex Inspection	Viewed binary + ASCII	Hexed.it	eicar.com.txt	hex sumpp of file

Activity	what have i done	tools used	input	output
Decode Content	Base64 decode	CyberChef	given encodede text	EICAR string revealed Answer 5

Indian Cyber Law and Compliance

1. Company Liability: Under Section 43A of the IT Act, 2000, can Helios Financial Services be legally penalised for failing to protect customer financial information that was leaked as a result of the malware infection? Explain the conditions under which liability is established.

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Under 43A section of 2000's IT act Helios Financial Services can be legally
penalised
    This is because company is leagally liable if it fails to protect the
sensitive data
    and Helios Financial Services is:
        1. handling customers's personal financial data
        2. the malware attack happened due to their negligence
        3. the data got reached which caused data loss

    therefore can be penalized if:
        1. they didnt had proper security controls
        2. they didn't had encryption or protection
        3. they didnt responded to the incident or poorly handled it
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2. Mandatory Reporting: According to the CERT-In Directions of April 2022, what are the exact steps and the maximum permitted timeframe within which Helios must report this cyber incident to the government authorities?

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As per CERT-In Directions, April 2022 (Section 2(1)), following are the
steps for constrainst for incident report
    must-to-have things
        1. Must report within 6 hours
        2. reporting to CERT-in
        3. Including following things
            a. what is the nature of the incident
            b. what systems are affected
            c. timeline of the incident
            d. (IOC) indicators of compromises like source, domains, hashes

    steps
        1. detection of incident
        2. collection of logs and evidences
        3. reporting to cert-in within 6 hours
        4. preserving logs till 180 days
```

Questions

Answer the following five questions:

1. Why is computing a cryptographic hash (such as SHA256) one of the very first steps in static malware analysis? What are the limitations of relying on hash-based detection alone, and how can attackers easily bypass it?

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Hashing gives unique identity to any data/file/text which makes it eaiser
to identify it and its integrety, it also allows quick lookups in threat
intelligence platforms like virustotal, we don't have to execute the file
while doing so.

limitations of hashing
    1. a slight change in data can change the whole hash
    2. can't detect new or unknown malwares
    3. it is uneffective against polymorphic or packed malware

How attackers bypass it
    1. they can slightly modify the file to change its hash
    2. they can add junk bytes
    3. they can use packers or encryptions
    4. they can generate plymorphics veriosn of malware
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2. Differentiate between static and dynamic malware analysis. Give one realistic scenario where static analysis alone would be insufficient and explain what dynamic analysis would reveal in that situation.

Aspects	Static Analysis	Dynamic Analysis
execution	we dont execute	we do execute
safety	safer	it requires a seperated enviroment like a virtual box
speed	faster	slower
visibiity	limited as it is not in execution	full runtime behaviour

```
When will static analysis fail
    1. the the malware is packed then static method will only show
encrypted or obfuscated code
    2. no meaning full string can be found by byte analysis

What synamic analysis reveals
    it release what the malware is actually doing in execution
    it can
        1. create files
        2. change registry
        3. can connect to remote server/machine
        4. process injection
```

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3. What are Indicators of Compromise (IOCs) and why are they central to a SOC analyst's workflow? Give at least four distinct types of IOCs and provide one example of each.

```
indicators of incident are the artifacts or evidences that indicates that a
system has been compromized

why are they central?
    1. they are used toe detect, investigate and respond to the threats
    2. helps in removing or blocking them
    3. can enable automatic operations in SOC tools
```

type	example
file hash	131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267 Answer 2
ip address	10.2.255.7
domain	malicious-site.com
File path	/tmp/malware.AppImage

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4. Explain the concepts of obfuscation and packing in the context of malware. Why do attackers use these techniques, and what tools or signs would help an analyst detect that a binary has been packed?

```
Obfuscation
    is a technique to hide code logic =, this is done to make analysis
difficult this can be done by threat actors to prevent detection in systems
or for intellectual property protections too

Packing
    it is the process of compression or encryption of executable file which
is used to wrap code to help it evade signature based detection
```

```
attackers use these techniques to
    1. evade antivirus detections
    2. prevention from reverse engineering
    3. to delay the analysis

we can detect is a binary has been packed by
    1. if it has very few radable string in hexdump
    2. file size is small but its behavious is complex
    3. has suspicious sections structure
    4. has high entropy

tools to detect
    1. Detect It Easy (DIE)
    2. PEStudio
    3. binwalk
```

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5. Imagine you are the SOC analyst for Helios Financial Services and your static analysis has confirmed that the suspicious attachment is malicious. Outline the immediate next three steps you would take, in order, and justify why each step is important.

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1. Isolate the affected system
    Disconnect from network
    Prevent malware spread
2. Collect and preserve evidence
    Gather logs, memory dump, file samples
    Maintain chain of custody
    Needed for investigation
3. Block IOCs across systems
    Block hash, IPs, domains in firewall/EDR
    Prevent reinfection
```